SILENT CONTEXT

Expert advice lives in chats and emails, and stays there.

The next person solves the same fault **from zero.**

Elisa's challenge: capture what field engineers and network experts learn, without giving those people extra work; tie it to the right network entities with its source and period of validity; and let an agent use it the next time a similar case appears.

Petri Korhonen · github.com/pkorrrhonen-cmd/silent-context · MIT

Veikko Vikkelä Field Engineer · 09:14
Changed the SFP on transmission node TRN-042. Link is up and alarms cleared, but traffic is still not coming back.

Erkki Expert Network Expert · 09:19
Check whether the traffic counters are actually increasing. If they stay at zero even though the interface is UP, it's probably the issue I've seen before.

Veikko Vikkelä Field Engineer · 09:21
Yes. Link UP. RX power looks normal. Counters basically zero.

Erkki Expert Network Expert · 09:22
That's it. Disable the interface, wait a little, then enable it again. Don't reboot the whole unit unless that fails.

Veikko Vikkelä Field Engineer · 09:26
Is this documented somewhere?

Erkki Expert Network Expert · 09:27
Not that I know of. We've just learned it over the years.

Teams excerpts · Elisa's example file, unchanged

THE ANSWER

Silent context turns those messages into claims an agent can use

READ AS WRITTEN

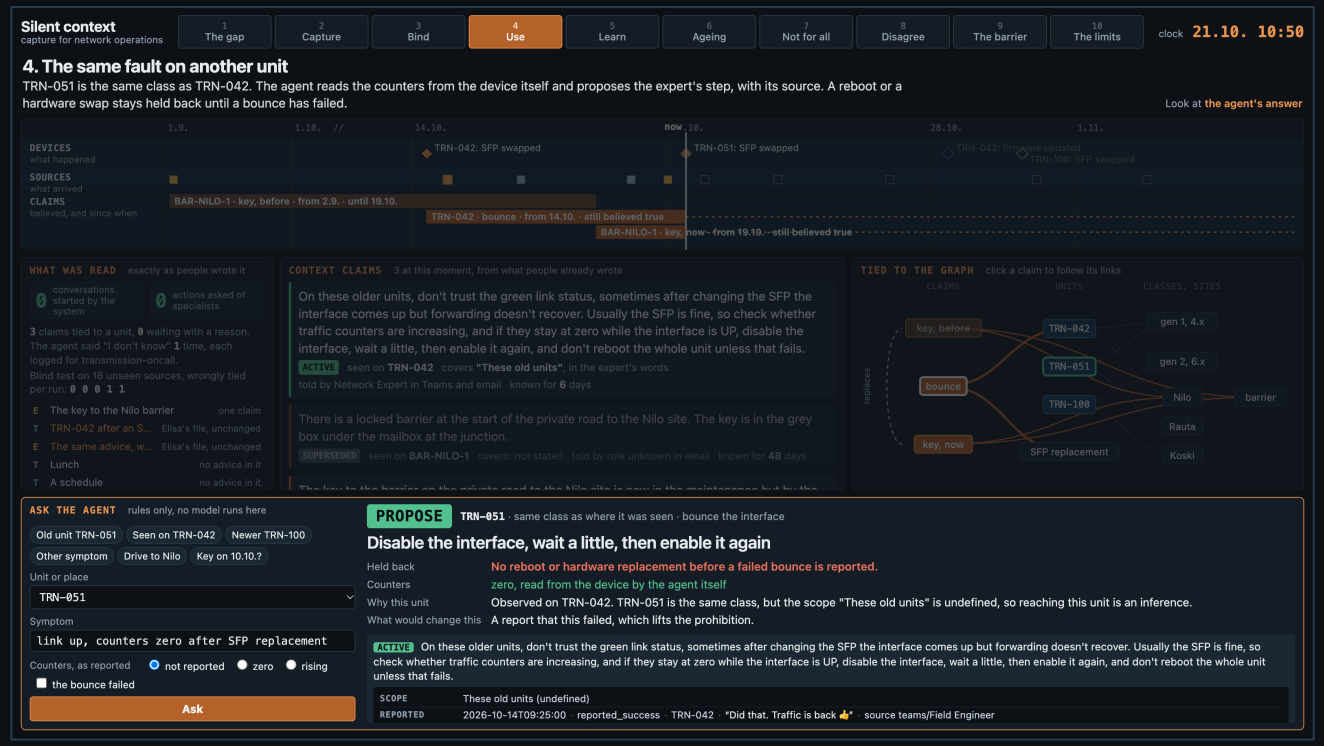
Teams threads and emails, exactly as they are. Nobody fills in anything for the system.

TIED TO THE NETWORK

A context claim on the unit, with its source, the expert's own words on which units it covers, and two clocks: when it holds in the world, and when we learned it.

ONE CALL FOR THE AGENT

`claims_for` answers with a next step and its sources: **propose**, **check first**, or **I don't know** with who to ask.



The page: ten scenes, no model and no key. Every answer on it is computed when asked, and the time axis shows what was known when.

CAPTURE AND BIND

The model reads; code binds, or waits with a reason

READ

One model call per thread reports what the text says. An identifier counts only if it is literally in the text.

BIND

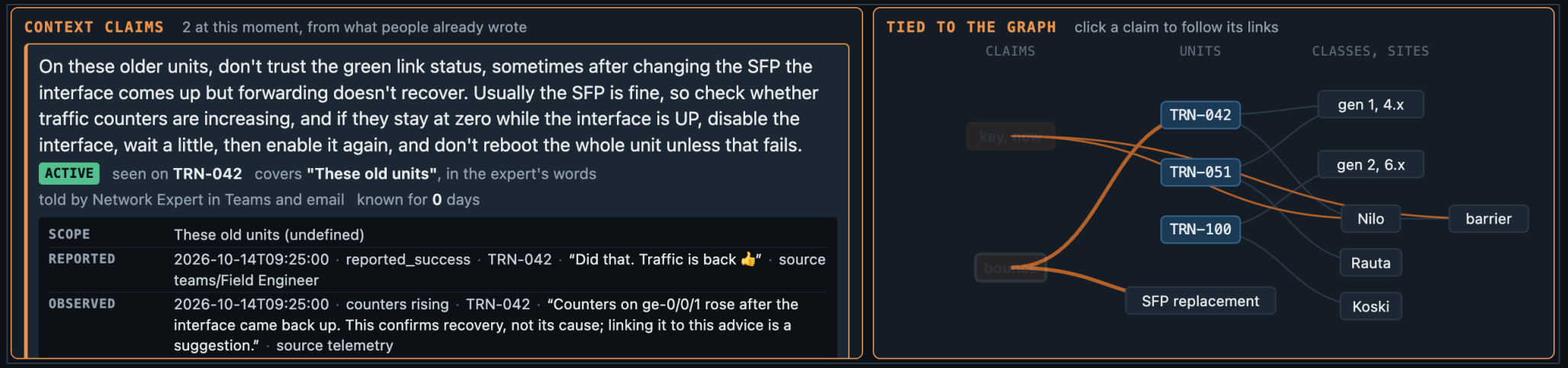
Code ties the claim to the graph. The model never picks an entity.

WAIT

An unclear binding waits with its reason: no unit named, several could fit, an unknown id, a contradicting context. It may wait forever. The system never asks anyone.

MERGE

One person's accounts become one claim with several sources. A relay is a source, not a telling.



THE DATA MODEL

A claim keeps its words, its scope and two clocks

WORDS

Claim text in the speaker's words, with “sometimes” and “usually” kept.

ENTITY AND SCOPE

The entity it was seen on, and apart from it the scope: wording verbatim, status confirmed or undefined.

`seen on TRN-042 · covers “These old units” (undefined)`

TWO CLOCKS

World time and information time apart. Unknown time stays unknown. No age limit is derived from “installed around 2019”.

`from 14.10. · still believed true`

EVIDENCE

Events with a date and a link: told, reported success, reported failure, disputed, superseded. Rendered from a template: no counts, no percentages, no figure per person.

STATUS

Computed at read time for a claim and a unit, never stored.

`ACTIVE · SUPERSEDED · DISPUTED`

THE GRAPH

Entities: units, classes, sites, a barrier, a procedure type and groups. Exported as graph.json and graph.cypher; the Cypher loads into Neo4j 5 with no edits.

THE AGENT

Three tiers from rules, one prohibition in code, no model

PROPOSE

A success is reported on this unit's class, no disagreement, and the claim's own check holds.

CHECK FIRST

Only told, in doubt, a guess, or disputed.

I DON'T KNOW

With the group to ask, and the gap logged.

PROPOSE **TRN-051** · same class as where it was seen · bounce the interface

Disable the interface, wait a little, then enable it again

Held back	No reboot or hardware replacement before a failed bounce is reported.
Counters	zero, read from the device by the agent itself
Why this unit	Observed on TRN-042. TRN-051 is the same class, but the scope "These old units" is undefined, so reaching this unit is an inference.
What would change this	A report that this failed, which lifts the prohibition.

The page's fourth scene: the same fault on TRN-051, the same class as TRN-042. The agent reads the counters from telemetry itself when a recent sample exists, proposes the expert's step with its source, and holds back a reboot or a hardware replacement until a failed bounce is reported.

What a person did, and the outcome, fill in from the device events. Nobody reports anything.

WHAT WAS MEASURED

Everything below runs with no key, from the recorded answers

154

tests pass

The data set's shape, the store's rules, extraction and binding against the truth set, the agent's acceptance cases, the page, the MCP wrapper.

11 / 11

truth set sources found

11 sources: merged 7/7, entity 4/4, scope 4/4, time 7/7, evidence 6/6.

1 of 16

confidently wrong, held-out binding set

16 synthetic sources the rule author has not read. Five repeats with the prompt unchanged: 0, 0, 0, 1, 1, the same file each time.

20 of 20

acceptance rows hold

The build plan's cases A0 to A11, written by app.accept from the recorded answers. A test keeps every computed row passing.

Passed

clean install from the README

git archive into an empty folder, .env with no values, every command in the README runs with no key, on 7.11.2026.

No edits

Cypher export loaded into Neo4j 5

5 claims, 14 entities, 14 evidence events and 9 sources as nodes, all ten edge types. One query answered the agent's question from the graph alone.

These numbers describe this synthetic data only. Not tested: extraction on real conversations, outcomes read from real chat, and the class binding on anything but this data.

ELISA'S QUESTION

Is the information fact? The answer is a claim's origin, not a score

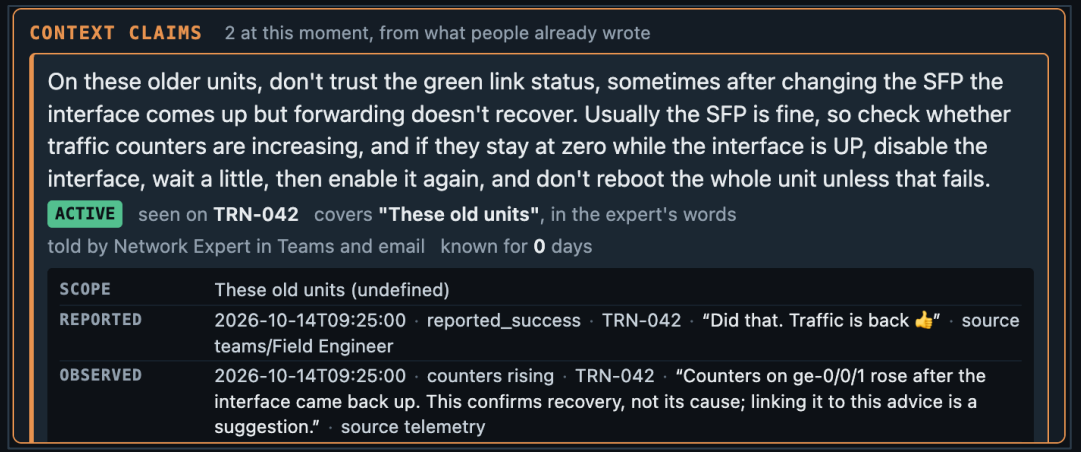
The answer is a claim's origin, what was reported about it, and when it was last confirmed. Not a score for the person.

WHY NO PERSON SCORES

- An employee assessment it was never meant to be.
- Personal data it does not need.
- A source that dries up once it is graded.

WHY NO COUNTS

“Confirmed twice” invites a trust the data does not carry. The events are listed; the reader counts if they want to.



One claim on the page: its words, where it was seen, which units it covers in the expert's own words, who told it and where, how long it has been known, and the evidence as events with their source.

WHAT IS REAL, WHAT IS SIMULATED

The mechanism is real; the network behind it is not

REAL

- Elisa's two files, unchanged; a test holds their checksums.
- The model's reading of every source, recorded once and replayed; not called at demo time.
- Binding, dating and merging, in code, live on every rebuild of the store.
- The agent: claims_for and access_for, rules and a template, no model.
- The exports: graph.json and graph.cypher, written from the store.

SIMULATED

- Every other source: nine threads and emails written for this demo.
- The dates: Elisa's files carry none, so a manifest gives each source a received time.
- The event stream and the telemetry; there is no network behind them.
- The graph: three units, two classes, three sites, a barrier, a procedure type, two groups.
- The demo clock and the channels. No real network, channel or person.

NOT BUILT, ON PURPOSE

- Asking the expert at the moment of need: the gap log names the group and the question; the asking is not built.
- Matching Elisa's own inventory graph; its schema is not known here.
- A semantic symptom match; symptoms are matched by plain words.
- Acting on the network: the agent proposes, a person acts.
- A mark-as-wrong button; first in the plan's cutting order.

WHAT COMES NEXT

Asking at the moment of need, and a record for autonomy

THE QUESTION AT THE MOMENT OF NEED

Elisa's own line: if the agent does not know, a person can ask Erkki. As a continuation of the gap log, a Teams message or a 20-second call when Erkki is driving away from the site. Not built, because synthetic data cannot show it is not extra work.

A QUESTION BUDGET

A question budget and situation conditions, so the asking never becomes the extra work it replaces.

AN AUTONOMY POLICY

Certainty is necessary, never sufficient. The log of proposed, done and outcome is the record that decision needs, and the demo already fills it from the device events.

FOR PRODUCTION AT ELISA

Schema matching against the inventory graph, channel connectors for Teams and email, and a semantic symptom match.

THE LIMITS, SAID PLAINLY

What this demo does not claim

01 The system never starts a conversation with a specialist.

02 It shows the origin of a claim and what was reported about it, not the truth.

03 It scores claims, not people: no personal assessment and no figure shown per person.

04 Telling apart which units a claim applies to, before someone narrows it down, rests on synthetic data.

05 Everything except Elisa's two example files is simulated.

06 Asking the expert at the moment of need is not built.

github.com/pkorrrhonen-cmd/silent-context MIT Petri Korhonen